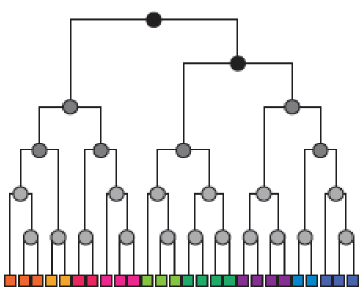
<u>Journal Club:</u>
Hierarchical Structure and the prediction of missing links in networks



Statistical Data Mining II

Rachael Hageman Blair

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LETTERS

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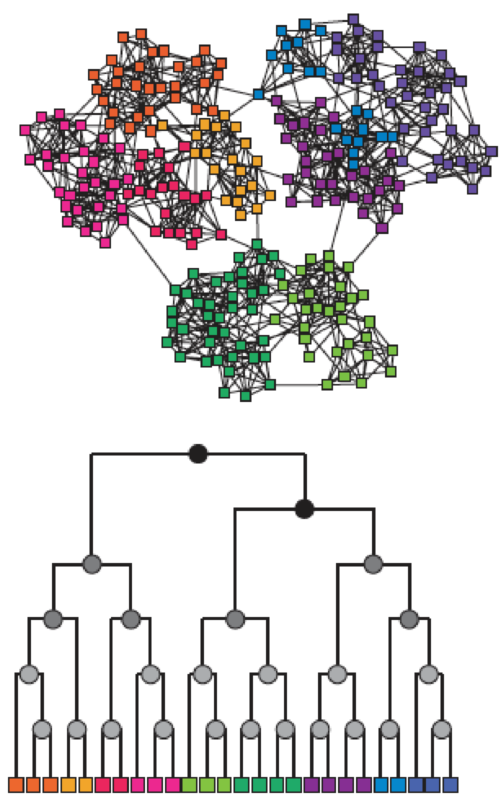

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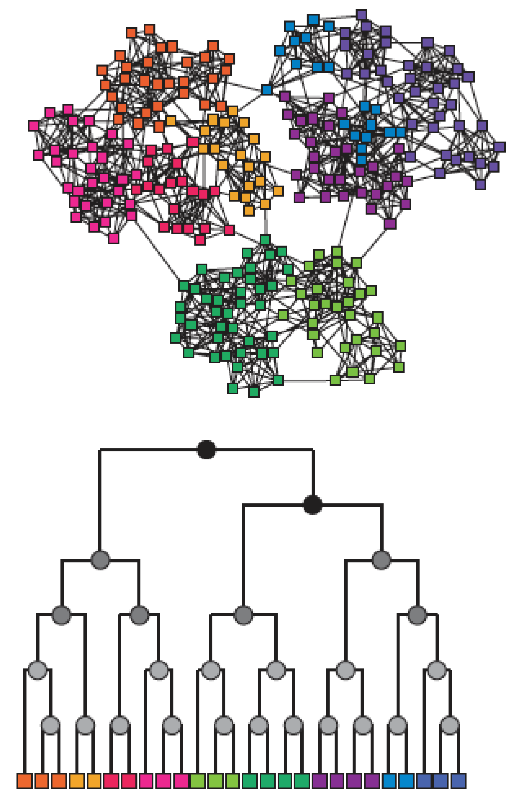
Aaron Clauset[1,3], Cristopher Moore[1,2,3] & M. E. J. Newman[3,4]



Motivation:

Link Prediction

Hierarchical Organization Principles in Networks.

Outline

- Motivation
- Introduction
- The Approach (Materials and Methods)
- Results
- Conclusion
- Afterthoughts… three questions

Motivation

- Link prediction in networks an open problem:
  examples: Biology, National Security, Social Networks.

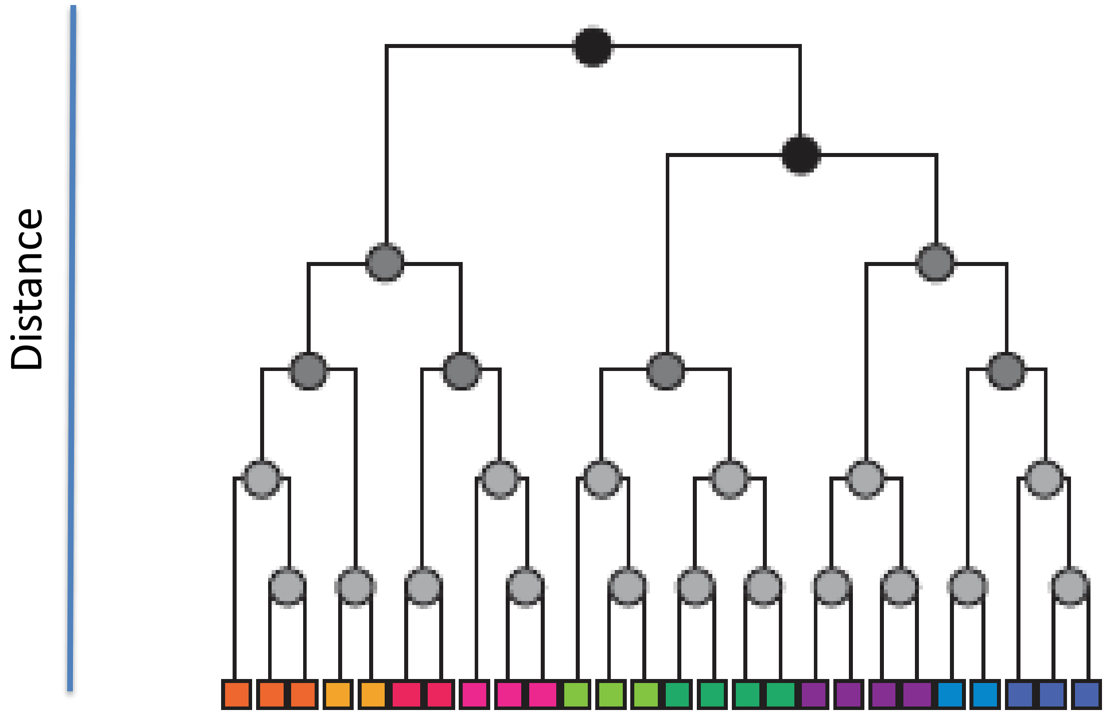
- What is meant by hierarchical organization?
  - Vertices divide into groups that further subdivide, and subdivide… etc. over multiple layers.

- Networks in general exhibit a hierarchical organization, which offers unique information, which cannot be extracted from the topology or measures on the topology.

- The hierarchical organization can be used as measure for link prediction.

Introduction

- Hierarchical structure can be represented by a dendrogram.



JC: Clauset

Introduction

- Node relatedness, some examples:

  let $G(i)$ and $G(j)$ be the neighbors of node i and node j.

- The probability of a connection between i and j, may depend on the degree of relatedness between the two.

  This can be modeled using:

  - Feature-based measures on a network (large-scale)

  - Using a PGM (not large-scale)

- One approach is to compute all possible relatedness measures and rank them. Those at the top of the list are the "most-likely".

Introduction

- Node relatedness, some examples:

  let $G(i)$ and $G(j)$ be the neighbors of node i and node j.

  - *Common neighbor* assigns $score(i,j) = |\Gamma(i) \cap \Gamma(j)|$, where $|\cdot|$ is the set cardinality.
  - *Adamic/Adar* computes similarity of features (neighbors) shared between objects (nodes) [1]. In our setting, the measure is given as: $\sum_{z \in \Gamma(i) \cap \Gamma(j)} \frac{1}{log|\Gamma(z)|}$, where z is a common neighbor.
  - *Jaccard's coefficient* denotes the probability that i and j share a common feature (neighbor) for a randomly selected feature (neighbor) that either i or j has, $score(i,j) = |\Gamma(i) \cap \Gamma(j)| / |\Gamma(i) \cup \Gamma(j)|$.
  - *Degree product* is the product of neighbors, $score(i,j) = |\Gamma(i)| \cdot |\Gamma(j)|$.
  - *Shortest path* is defined simply as one divided by the length of the shortest path between nodes i and j and zero if there is no path between them.

The Approach

- Model the connection between vertices to depend on the degree of relatedness.

Hierarchical random graph

- similar in spirit to tree-based models.
- Each internal node r of the dendrogram has a probability p_r, and connect the vertices for which r is the lowest common ancestor independently with probability p_r.
- A sampling implementation, which takes into account uncertainty.

The Approach: details

See Supplement

- Notation:

  Let G be a graph with n vertices.

  Let D be a dendrogram with leaves corresponding to the vertices.

  p_r a probability associated with each internal node r .

  Given two vertices i and j of G the probability p_{ij} that they are connected by an edge is $p_{ij} = p_r$, where r is the lowest common ancestor in D.

- Hierarchical Random Graph – defined by $\left(D, \{p_r\}\right)$.

The Approach: details

How to fit a hierarchical random graph to data

- How do we find he random graph or graphs that best fit a network G?
- Assumption, all hierarchical graphs are equally likely, *a priori*. posterior - > likelihood.
- Let E_r be the number of edges in G whose endpoints have r as their lowest common ancestor in D, and let L_r and R_r, be the leaves in the left and right subtrees rooted at r.
- The likelihood of the hierarchical random graph is:

$$L(D,\{p_r\}) = \prod_{r \in D} p_r^{E_r} (1 - p_r)^{L_r R_r - E_r}$$

The Approach: details

- For a fixed dendrogram:

$$\bar{p}_r = \frac{E_r}{L_r R_r}$$

is the fraction of edges between two subtrees of r that actually appear in the graph G.

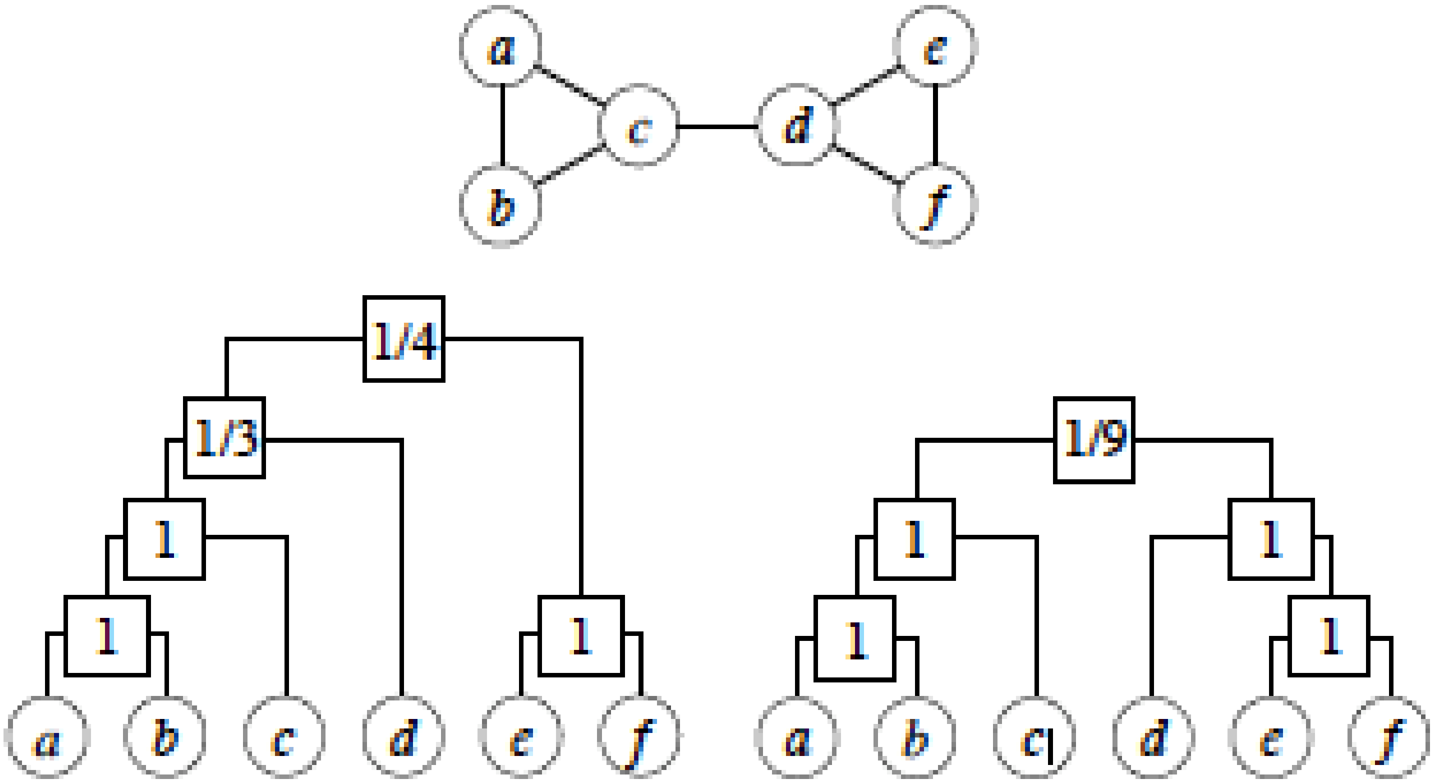
- The likelihood of the dendrogram evaluated at this maximum is:

$$L(D) = \prod_{r \in D} \left[\bar{p}_r^{\,\bar{p}_r} \left(1 - \bar{p}_r\right)^{1-\bar{p}_r} \right]^{L_r R_r}$$

which can be log transformed for simplicity (not shown).

JC: Clauset

The Approach: details

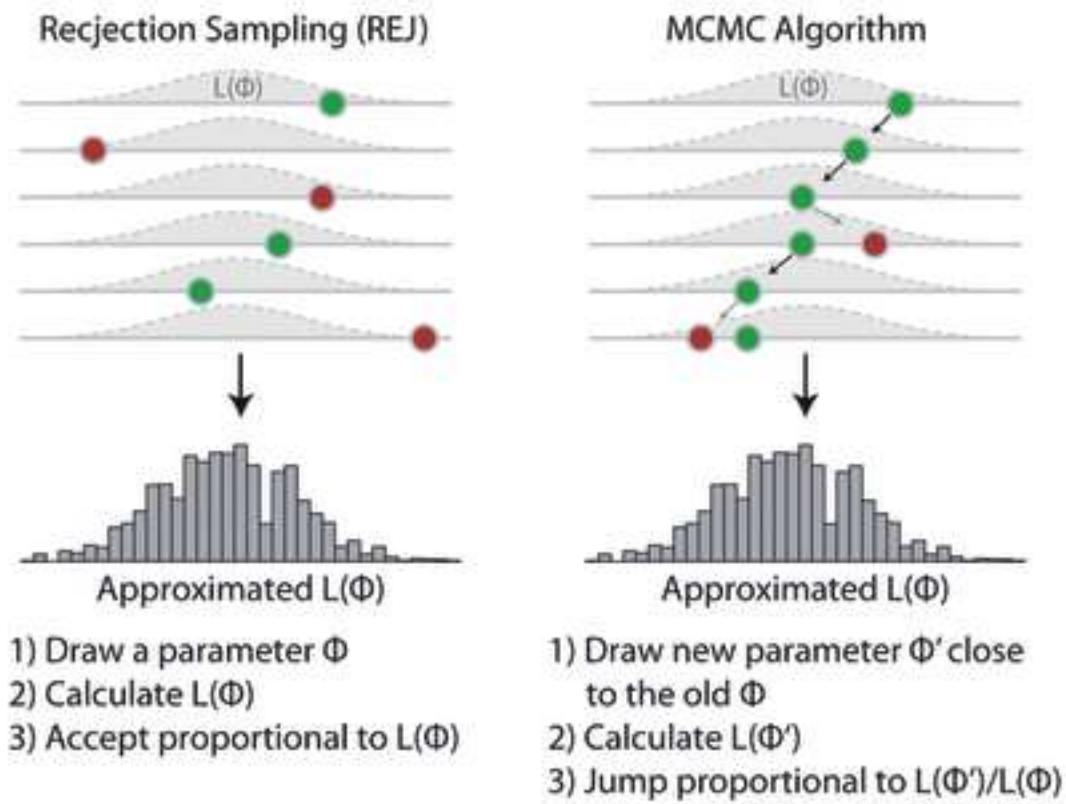


$$L(D) = (1/3)(2/3)^2 \times (1/4)(3/4)^6$$
$$= 0.00165$$

$$L(D) = (1/9)(8/9)^8$$
$$= 0.0433$$

The Approach: details

- We need a way to sample dendrograms … explore the posterior model space.



Approach: Metropolis Hastings

The Approach: details

- We need a way to sample dendrograms … explore the posterior model space.

Recall: Metropolis-Hastings Search Strategy

$$\text{Acceptance} = \min\left\{ \frac{P(G \mid D_{new})\, P(D_{new})\, Q(D_{curr} \mid D_{new})}{P(G \mid D_{curr})\, P(D_{curr})\, Q(D_{new} \mid D_{curr})},\ 1 \right\}$$

**We need a way to move between dendrograms, “a proposal”.

The Approach: details

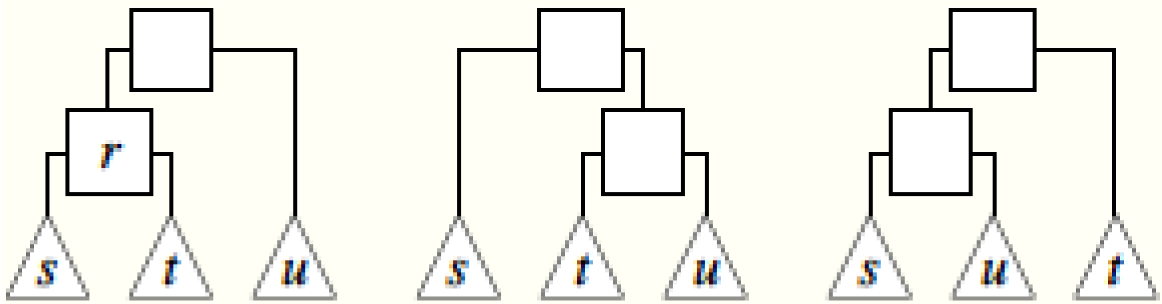


FIG. 2: Each internal node r of the dendrogram has three associated subtrees s, t, and u, which can be placed in any of three configurations. (Note that the topology of the dendrogram depends only on the sibling and parent relationships; the order, left to right, in which they are depicted is irrelevant).

The Approach: details

- In their implementation, the new dendrogram:

$$D_{old} \rightarrow D_{new},$$

is accepted if,

$$\mathrm{D}\log L = \log(L(D_{new})) - \log(L(D_{old})) > 0$$

i.e., the new dendrogram is more likely.

Similar in spirit to a greedy search, less like Metropolis-Hastings.

The Approach: details

In practice, … "there are typically many dendrograms with roughly equal likelihoods, which reinforces our contention that it is important to sample the distribution of dendrograms rather than merely focusing on the most likely one."

Applications

Three very different networks:

- Treponema pallidum metabolic network (bacteria → disease).

- Food web of grassland species.

- Terrorist Network.

Valdis E. Krebs
orgnet.com

This paper looks at the difficulty in mapping covert networks. Analyzing networks after an event is fairly easy for prosecution purposes. Mapping covert networks to prevent criminal activity is much more difficult. We examine the network surrounding the tragic events of September 11th, 2001. Through public data we are able to map a portion of the network centered around the 19 dead hijackers. This map gives us some insight into the terrorist organization, yet it is incomplete. Suggestions for further work and research are offered.

The Approach: details

"Validation of approach"

1. use the sampled dendrograms to produce new networks (G), different from the original, but retaining the same hierarchical structure.
2. Hierarchical organization.
3. Link Prediction.

The Approach: details

Appendix C: Resampling from the hierarchical random graph

The procedure for resampling from the hierarchical random graph is as follows.

1. Initialize the Markov chain by choosing a random starting dendrogram.

2. Run the Monte Carlo algorithm until equilibrium is reached.

3. Sample dendrograms at regular intervals thereafter from those generated by the Markov chain.

4. For each sampled dendrogram D, create a resampled graph G' with n vertices by placing an edge between each of the $n(n-1)/2$ vertex pairs (i, j) with independent probability $p_{ij} = \overline{p}_r$, where r is the lowest common ancestor of i and j in D and $\overline{p}_r$ is given by Eq. (B2). (In principle, there is nothing to prevent us from generating many resampled graphs from a dendrogram, but in the calculations described in this paper we generate only one from each dendrogram.)

Results

Table 1 | Comparison of original and resampled networks

Network	$\langle k \rangle_{real}$	$\langle k \rangle_{samp}$	C_{real}	C_{samp}	d_{real}	d_{samp}
T. pallidum	4.8	3.7(1)	0.0625	0.0444(2)	3.690	3.940(6)
Terrorists	4.9	5.1(2)	0.361	0.352(1)	2.575	2.794(7)
Grassland	3.0	2.9(1)	0.174	0.168(1)	3.29	3.69(2)

Statistics are shown for the three example networks studied and for new networks generated by resampling from our hierarchical model. The generated networks closely match the average degree $\langle k \rangle$, clustering coefficient C and average vertex–vertex distance d in each case, suggesting that they capture much of the structure of the real networks. Parenthetical values indicate standard errors on the final digits.

Results

Appendix C: Resampling from the hierarchical random graph

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Results

Appendix D: Predicting missing connections

Our algorithm for using hierarchical random graphs to predict missing connections is as follows.

1. Initialize the Markov chain by choosing a random starting dendrogram.
2. Run the Monte Carlo algorithm until equilibrium is reached.
3. Sample dendrograms at regular intervals thereafter from those generated by the Markov chain.
4. For each pair of vertices i, j for which there is not already a known connection, calculate the mean probability $\langle p_{ij} \rangle$ that they are connected by averaging over the corresponding probabilities p_{ij} in each of the sampled dendrograms D.
5. Sort these pairs i, j in decreasing order of $\langle p_{ij} \rangle$ and predict that the highest-ranked ones have missing connections.

In general, we find that the top 1% of such predictions are highly accurate. However, for large networks, even the top 1% can be an unreasonably large number of candidates to check experimentally. In many contexts, researchers may want to

Results

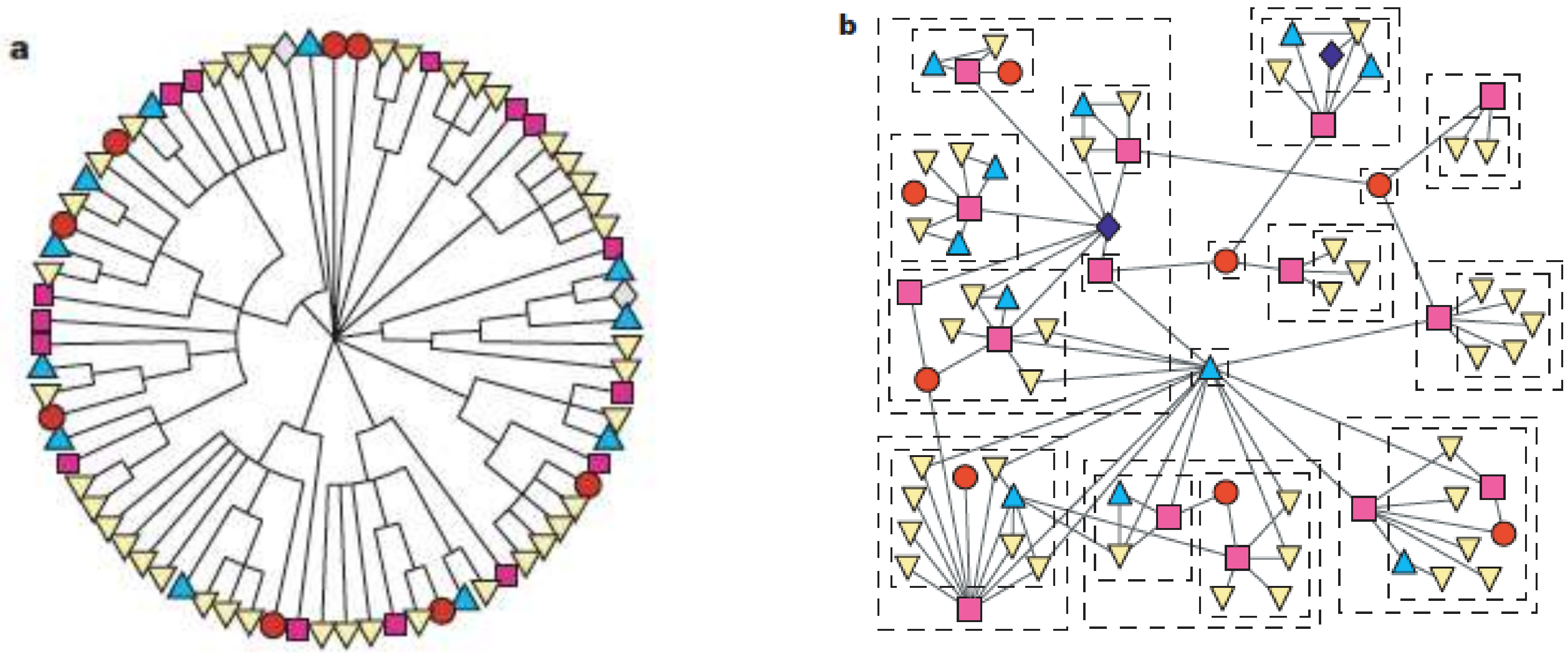

Figure 2 | Application of the hierarchical decomposition to the network of grassland species interactions. **a,** Consensus dendrogram reconstructed from the sampled hierarchical models. **b,** A visualization of the network in which the upper few levels of the consensus dendrogram are shown as boxes around species (plants, herbivores, parasitoids, hyperparasitoids and hyper-hyperparasitoids are shown as circles, boxes, down triangles, up triangles and diamonds, respectively). Note that in several cases a set of parasitoids is grouped into a disassortative community by the algorithm, not because they prey on each other but because they prey on the same herbivore.

Results

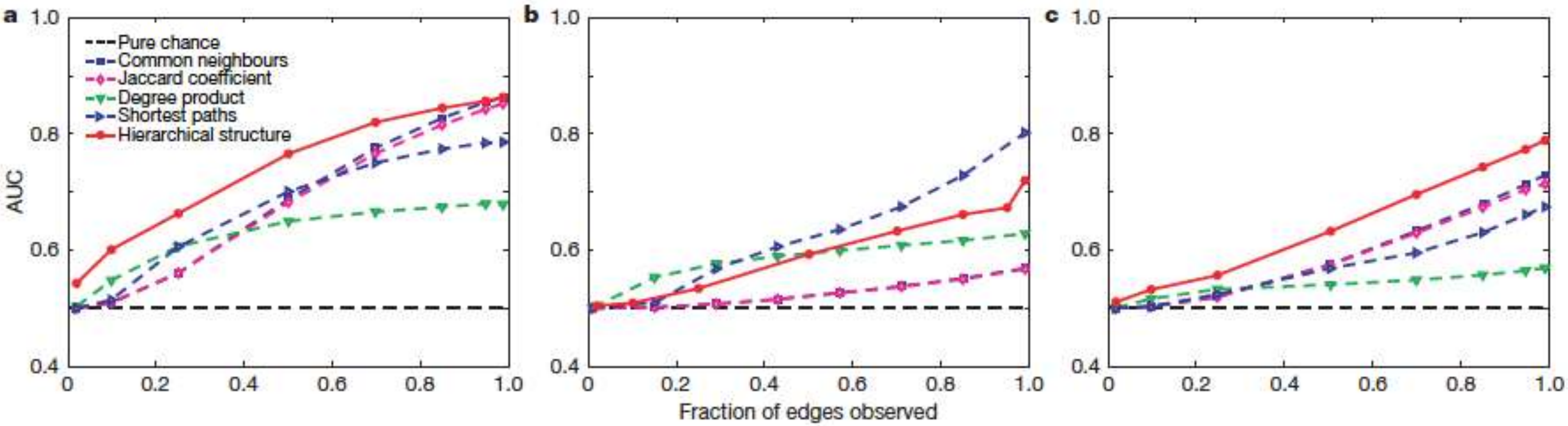

Figure 3 | Comparison of link prediction methods. Average AUC statistic—that is, the probability of ranking a true positive over a true negative—as a function of the fraction of connections known to the algorithm, for the link prediction method presented here and a variety of previously published methods. **a**, Terrorist association network; **b**, *T. pallidum* metabolic network; **c**, grassland species network.

Conclusion

- Hierarchical organization captures the important structural features of a network.
- Many dendrograms are close in probably, sampling and summarizing through consensus or averaging, will better identify features and organization.
- Some evidence of link prediction value, against certain metrics.

Afterthoughts…. Three questions

1. Is the method scalable to large networks?

2. How do you summarize dendrograms?

3. What happens if the network is sparse?

4. Are there better ways to test accuracy in link prediction.